

MODULE 08

Pelvis & Hip CT: Fracture Classification Systems

Young-Burgess · Tile (AO) · Garden · Pauwels · Denis · Pipkin · Judet-Letournel

1.0 AMA PRA Cat 1

Pelvis

CME Article Review

LEARNING OBJECTIVES

1. Apply the Young-Burgess classification to pelvic ring fractures (lateral compression, anteroposterior compression, vertical shear, and combined-mechanism patterns).
2. Apply the Tile (AO) classification as a complementary stability-based framework for pelvic ring injury.
3. Grade femoral neck fractures using the Garden (displacement) and Pauwels (fracture-line orientation) systems.
4. Apply the Denis classification to sacral fractures (Zones I–III) and recognize their associated neurologic risk.
5. Classify femoral head fractures associated with hip dislocation using the Pipkin system.
6. Identify complex acetabular fracture patterns using the Judet-Letournel classification, including elementary versus associated patterns and the both-column "spur sign."

This module is produced by the LucidMSK Editorial Board for educational purposes only. Upon completion of all content and a post-test score $\geq 70\%$, participants may claim the stated 1.0 AMA PRA Category 1 Credit(s)[™]. No commercial support was received.

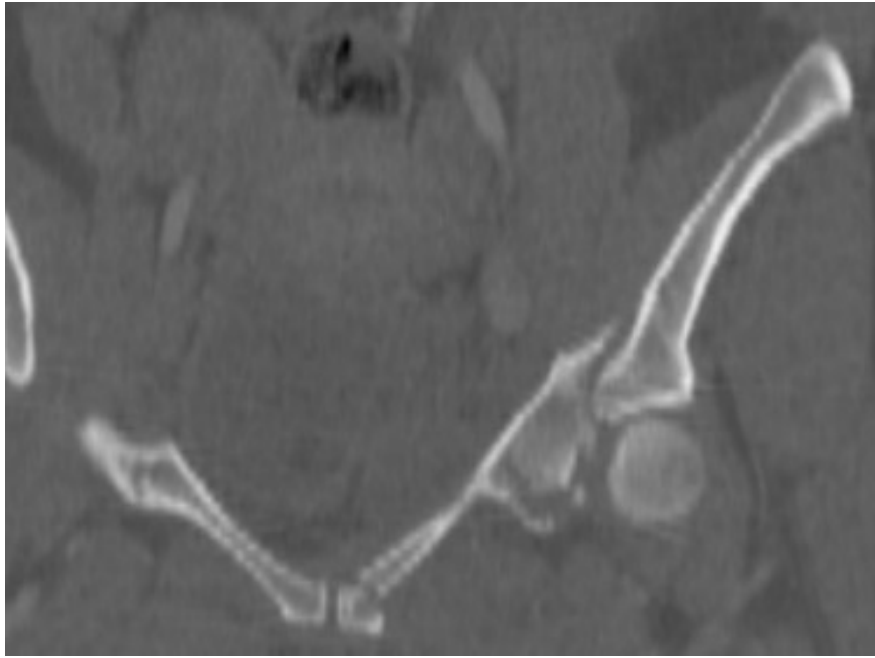
1. Young-Burgess Classification — Pelvic Ring Fractures

The Young-Burgess classification organizes pelvic ring fractures by the mechanism of injury into four families: lateral compression (LC), anteroposterior compression (APC), vertical shear (VS), and combined mechanism (CM). The system is valued in the trauma setting because mechanism correlates directly with both mechanical stability and hemorrhage risk, allowing rapid triage decisions before definitive fixation planning.

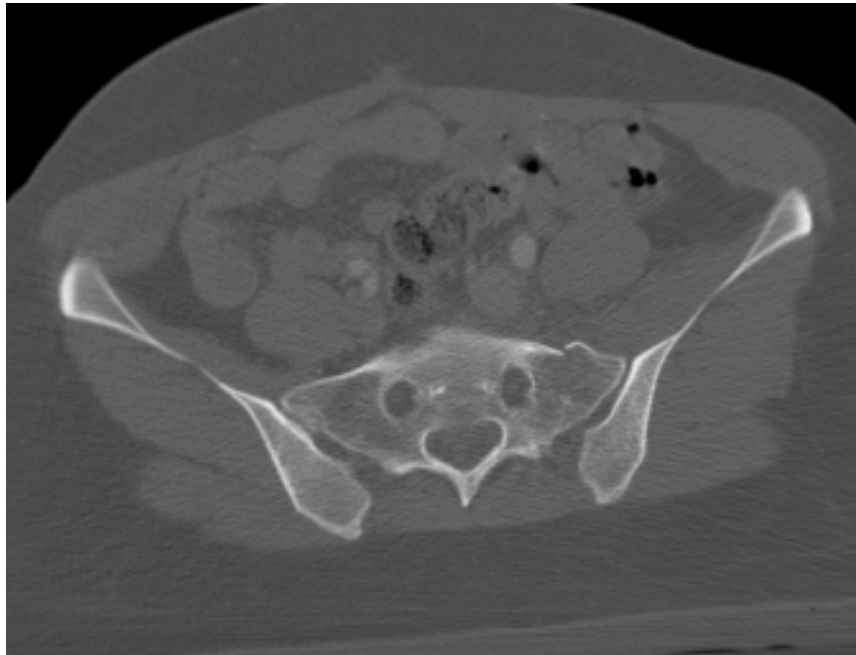
Young-Burgess Classification — Pelvic Ring Fractures

Type	Subtype	Key Features	Stability / Hemorrhage Risk
Lateral Compression (LC)	I	Sacral buckle (impaction) fracture; ipsilateral horizontal pubic rami fracture	Stable; low hemorrhage risk
Lateral Compression (LC)	II	LC-I plus an ipsilateral posterior iliac wing ("crescent") fracture	Partially unstable
Lateral Compression (LC)	III	LC-II plus a contralateral open-book (APC-type) injury — "windswept pelvis"	Highly unstable; major hemorrhage risk
Anteroposterior Compression (APC)	I	Symphyseal diastasis < 2.5 cm; SI ligaments intact	Stable
Anteroposterior Compression (APC)	II	Diastasis > 2.5 cm; anterior SI ligaments torn, posterior SI ligaments intact ("open book")	Partially unstable; moderate hemorrhage risk
Anteroposterior Compression (APC)	III	Complete SI joint disruption — both anterior and posterior SI ligaments torn	Most unstable; major hemorrhage risk
Vertical Shear (VS)	—	Vertical displacement through the SI joint and/or sacrum (Malgaigne pattern)	Highly unstable; major hemorrhage risk

Beyond classification, several specific structures should be deliberately interrogated on every pelvic trauma CT regardless of the dictated indication: the pubic symphysis (diastasis width), both SI joints (symmetry and widening), the sacral ala bilaterally (buckle fractures are subtle and easily overlooked on axial images alone), and the posterior iliac wing for a crescent fragment. LC and APC injuries are distinguished largely by the direction of pubic rami fracture displacement and the presence or absence of symphyseal diastasis, while VS injuries are recognized by vertical malalignment best appreciated on coronal reformats, or by comparing the relative heights of the two hemipelvises.



Axial CT demonstrating pelvic ring disruption with widening of the anterior ring — an open-book-type pattern.



Axial CT through the pelvic ring in a separate trauma patient, for comparison of normal versus injured ring contour.

PITFALL

APC-III and vertical shear patterns carry the highest risk of life-threatening hemorrhage from injury to the internal iliac vessels. These patterns warrant urgent communication to the trauma team, since emergent pelvic binder placement and angioembolization may be needed before imaging is even fully reviewed in a stable, unhurried setting.

REFERENCES

- Young JW, Burgess AR, Brumback RJ, Poka A. Pelvic fractures: value of plain radiography in early assessment and management. *Radiology* 1986.
- Young JW, Burgess AR. *Radiologic Management of Pelvic Ring Disruptions*. Urban & Schwarzenberg, 1987.

2. Pelvic Ring Stability — Tile (AO) Classification

The Tile (AO) classification offers a complementary, stability-first framework to Young-Burgess. Where Young-Burgess organizes injuries by mechanism, Tile organizes them directly by the degree of posterior arch disruption and resultant mechanical stability — a structure that maps cleanly onto surgical decision-making and is often the language used directly by orthopedic trauma teams.

Tile (AO) Classification — Pelvic Ring Stability

Type	Posterior Arch Status	Stability
A	Intact posterior arch	Stable — typically isolated anterior or avulsion-type injuries
B	Partially disrupted posterior arch (rotationally unstable, vertically stable)	Partially unstable — corresponds broadly to LC and APC-II patterns
C	Completely disrupted posterior arch (rotationally and vertically unstable)	Completely unstable — corresponds broadly to APC-III and vertical shear patterns

CLINICAL TIP

Young-Burgess and Tile are not competing systems but complementary languages for the same injury — using both in a report (e.g. "APC-III / Tile C pattern with complete posterior arch disruption") communicates clearly regardless of which system the receiving surgeon prefers, and reinforces the stability assessment that actually drives management.



Axial CT showing a sacral fracture component contributing to posterior arch disruption.

REFERENCES

- Tile M. Pelvic ring fractures: should they be fixed? *J Bone Joint Surg Br* 1988.

3. Femoral Neck Fractures — Garden & Pauwels Classifications

Femoral neck fractures are graded with two complementary systems. The Garden classification is based on the degree of displacement and is the primary prognostic system used to estimate risk of avascular necrosis (AVN) of the femoral head, since displacement disrupts the retinacular vessels supplying the femoral head. The Pauwels classification instead describes the orientation of the fracture line relative to horizontal, adding biomechanical context about the shear versus compressive forces acting across the fracture.

Garden Classification — Femoral Neck Displacement

Stage	Description	AVN Risk	Typical Management
I	Incomplete or impacted fracture; valgus impaction; no true displacement	~5%	Fixation (cannulated screws) or, in select cases, conservative management
II	Complete fracture; no displacement; trabecular alignment maintained	~10%	Fixation with cannulated screws
III	Complete fracture; partial displacement; trabecular misalignment between head and neck	~25–35%	Fixation in younger patients; arthroplasty favored in elderly patients
IV	Complete fracture; full displacement; femoral head freely rotated, trabeculae of head realign with the acetabulum	~35–50%	Total hip arthroplasty or hemiarthroplasty in elderly patients; fixation considered in younger patients



Subtle subcapital femoral neck fracture line on radiograph — early Garden-stage injury without gross displacement.



Femoral neck fracture on CT demonstrating the fracture line at the head-neck junction.

Pauwels Classification — Fracture Line Orientation and AVN Risk

Type	Fracture Angle from Horizontal	Force Pattern	AVN / Non-union Risk
I	< 30° (near horizontal)	Predominantly compressive — most mechanically stable	Lowest
II	30–50° (intermediate)	Mixed compressive and shear forces	Moderate
III	> 50° (near vertical)	Predominantly shear — least stable construct	Highest non-union and AVN risk



Coronal CT showing a displaced femoral neck fracture with varus angulation and cortical disruption — a high Garden-stage, high Pauwels-angle pattern.



Post-fixation radiograph showing percutaneous guide pins/screws traversing the femoral neck — the typical fixation construct for a stable Garden I–II pattern.

PEARL

Garden stage is the primary prognostic system for AVN risk and guides the choice between fixation and arthroplasty, especially in older patients. Pauwels adds biomechanical context on top of Garden — use both together rather than either in isolation. In select young patients with a Pauwels III fracture, a valgus-producing intertrochanteric osteotomy can convert shear forces into compressive forces, improving healing potential.

REFERENCES

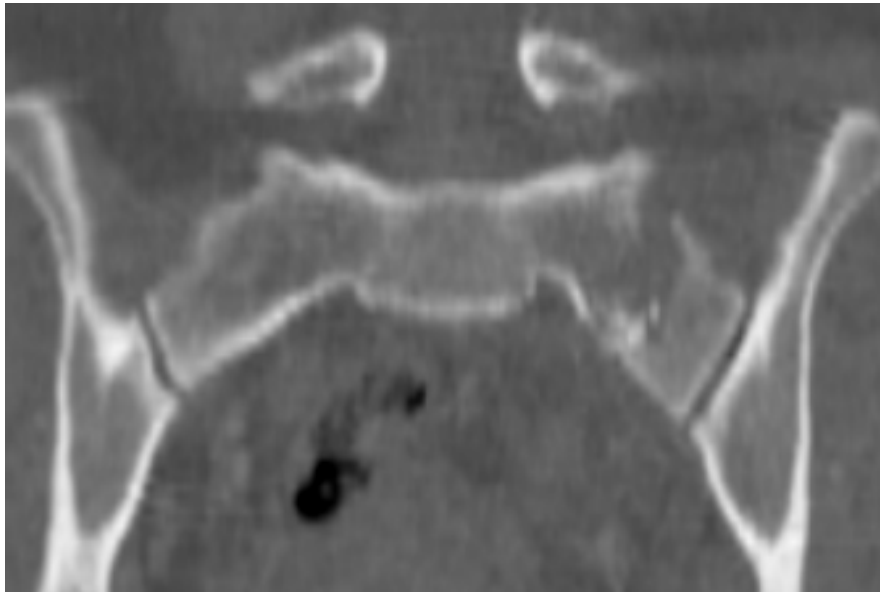
- Garden RS. Low-angle fixation in fractures of the femoral neck. *J Bone Joint Surg Br* 1961.
- Pauwels F. Der Schenkelhalsbruch, ein mechanisches Problem. *Z Orthop Chir* 1935 (biomechanical classification, widely cited in translation).

4. Sacral Fractures — Denis Classification

The Denis classification divides the sacrum into three zones based on the relationship of the fracture line to the neural foramina, and neurologic injury risk rises sharply and predictably from Zone I to Zone III — making zone identification one of the most clinically consequential pieces of information in any sacral fracture report.

Denis Classification — Sacral Fracture Zones

Zone	Location	Neurologic Deficit Rate	Typical Nerve Involvement
I (Alar)	Lateral to the neural foramina	~6%	L5 nerve root; the most common zone overall
II (Foraminal)	Through the sacral neural foramina	~28%	Unilateral sacral nerve root injury, typically
III (Central canal)	Medial to the foramina, involving the spinal canal	~57%	Cauda equina; bilateral root injury; bladder, bowel, and sexual dysfunction risk



Axial CT showing a sacral fracture line — assess its position relative to the neural foramina to assign the Denis zone.

PITFALL

A transverse sacral fracture spanning multiple zones (the "U-type" or "H-type" pattern when combined with bilateral vertical fractures) represents a particularly unstable and high-risk variant — always explicitly assess for a transverse component, not just vertical zone involvement, since it is easily overlooked on axial images alone and is better appreciated on sagittal reconstructions.

REFERENCES

- Denis F, Davis S, Comfort T. Sacral fractures: an important problem. *Clin Orthop Relat Res* 1988.

5. Femoral Head Fractures — Pipkin Classification

Femoral head fractures associated with hip dislocation — most commonly posterior dislocation — are classified by the Pipkin system based on fragment location relative to the fovea capitis and the presence of associated femoral neck or acetabular fractures. The distinction between fragments above versus below the fovea is the single most important branch point, since it determines whether the weight-bearing articular surface itself is involved.

Pipkin Classification — Femoral Head Fractures

Type	Description	Prognosis
I	Fracture fragment inferior to the fovea capitis — non-weight-bearing surface	Better prognosis
II	Fracture fragment superior to the fovea capitis — weight-bearing articular dome involved	Worse prognosis than Type I
III	Femoral head fracture with an associated femoral neck fracture	High AVN risk — disrupts retinacular blood supply
IV	Femoral head fracture with an associated acetabular fracture	Complex; combined reconstruction often needed

REFERENCES

• Pipkin G. Treatment of grade IV fracture-dislocation of the hip. *J Bone Joint Surg Am* 1957.

6. Acetabular Fractures — Judet-Letournel Classification

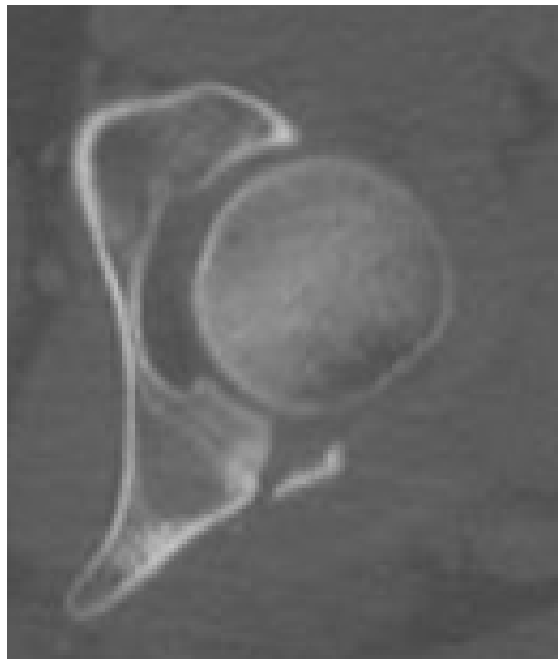
The Judet-Letournel system remains the standard classification for acetabular fractures, organizing patterns into five elementary (simple) types and five associated (combined, more complex) types. Elementary fractures involve a single column or wall; associated fractures combine two or more elementary patterns and account for a substantial share of operative acetabular fractures encountered in practice. Correct classification depends on systematic assessment of both acetabular columns (anterior and posterior) and both walls on multiplanar CT with 3D reformats.

Judet-Letournel — Elementary (Simple) Patterns

Pattern	Description
Posterior wall	Isolated fracture of the posterior acetabular rim, often with posterior hip dislocation — the most common elementary pattern
Posterior column	Fracture through the ischium and posterior ilium, separating the posterior column from the rest of the innominate bone
Anterior wall	Isolated fracture of the anterior acetabular rim; uncommon in isolation
Anterior column	Fracture extending from the iliac crest or anterior iliac spine through the anterior acetabulum
Transverse	Fracture line crossing both columns horizontally, dividing the acetabulum into superior and inferior fragments

Judet-Letournel — Associated (Complex) Patterns

Pattern	Description
T-shaped	Transverse fracture combined with a vertical stem through the obturator foramen/ischiopubic region, creating a "T" configuration
Posterior column + posterior wall	Combined posterior column fracture with an additional posterior wall fragment
Transverse + posterior wall	A transverse fracture combined with a posterior wall fragment — a common pattern after posterior dislocation
Anterior column/wall + posterior hemitransverse	Anterior column or wall fracture combined with a posterior fracture line that does not cross the full acetabulum
Both-column	All articular segments detached from the axial skeleton ("floating acetabulum"); pathognomonic "spur sign" — an isolated iliac wing fragment posterior to the fracture line; the most common complex pattern encountered in practice



Axial CT showing a posterior acetabular wall fracture fragment displaced relative to the femoral head — the classic elementary posterior wall pattern.

PITFALL

A posterior wall fragment involving more than 50% of the wall implies posterior hip instability after reduction and typically requires surgical fixation; fragments in the 25–40% range are borderline and may warrant a CT-based stress exam to determine whether fixation is needed. Always estimate and state the percentage of wall involvement explicitly rather than describing the fragment only qualitatively.

Reporting acetabular fractures benefits from a consistent structure: state which column(s) and wall(s) are involved, whether the fracture is elementary or associated, the direction and degree of any femoral head subluxation or dislocation, articular step-off or gap at the weight-bearing dome, and the presence of intra-articular fragments. This structure maps directly onto the information orthopedic trauma surgeons need for operative planning, regardless of

which specific named pattern ultimately applies.

REFERENCES

- Letournel E. Acetabulum fractures: classification and management. *Clin Orthop Relat Res* 1980.
- Judet R, Judet J, Letournel E. Fractures of the acetabulum: classification and surgical approaches for open reduction. *J Bone Joint Surg Am* 1964.

7. Additional Pelvic and Femoral Trauma Patterns

Two additional patterns are worth specific recognition alongside the major classification systems above, since they appear in pelvic and proximal femoral trauma workups and have distinct management implications.

Apophyseal Avulsion Fractures (Skeletally Immature Patients)

In adolescent athletes with open growth plates, forceful muscle contraction can avulse an unfused apophysis rather than producing a soft-tissue strain, since the apophyseal physis is biomechanically weaker than the adjacent tendon. The anterior superior iliac spine (ASIS, sartorius origin), anterior inferior iliac spine (AIIS, rectus femoris origin), and ischial tuberosity (hamstring origin) are the classic pelvic avulsion sites. Recognizing an open growth plate adjacent to a displaced bony fragment is the key to correctly identifying this entity rather than mistaking it for an aggressive lesion or unrelated trauma.



Frontal pelvis radiograph in a skeletally immature patient showing a right-sided apophyseal avulsion fracture at the ASIS/AIIS region, with an open growth plate confirming skeletal immaturity.

CLINICAL TIP

Apophyseal avulsions occur almost exclusively in adolescents with open growth plates — always confirm physeal status before committing to this diagnosis, and do not mistake the normal apophyseal lucency for a fracture, or a true avulsion fragment for a normal unfused apophysis, without correlating both sides and the clinical history of acute pain during forceful activity.

Atypical Femur Fracture

Atypical femur fractures occur in the subtrochanteric region or femoral shaft and are strongly associated with prolonged bisphosphonate (or other antiresorptive) therapy. The hallmark imaging feature is focal lateral cortical thickening ("beaking") at the fracture site, often with a transverse or short oblique fracture line and minimal comminution — a pattern distinct from a typical high-energy femoral shaft fracture. Contralateral femur imaging is recommended once an atypical fracture is identified, since these injuries are frequently bilateral, and an asymptomatic but radiographically similar stress reaction can be present on the other side.



Radiograph showing lateral cortical thickening ("beaking") of the proximal femoral shaft, the classic appearance of an atypical (bisphosphonate-associated) femur fracture.

REFERENCES

- Shane E et al. Atypical subtrochanteric and diaphyseal femoral fractures: second report of a task force of the ASBMR. *J Bone Miner Res* 2014.
- Rossi LM et al. Imaging of pediatric pelvic avulsion fractures. *Pediatr Radiol* 2015.